

RYNGE, Sweden

WMO: 026090

Lat: 55.4742N

Lon: 13.6683E

Elev: 164

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.3	18.6	8.8	8.7	16.7	13.0	10.7	21.0	31.0	39.3	29.0	39.0	5.8	0	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	77.8	65.0	75.0	63.7	72.5	62.8	67.4	74.1	65.6	71.7	64.3	70.0	9.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	92.7	70.7	63.3	87.7	68.6	61.9	83.2	67.1	32.0	74.3	30.6	72.0	29.5	70.2	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	22.3	20.0	DB	7.4	82.3	7.0	2.9	2.3	84.5	-1.8	86.2	-5.7	87.8	-10.9	90.0	(4)
(5)			WB	6.8	69.9	7.0	2.1	1.7	71.4	-2.4	72.7	-6.3	73.9	-11.4	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	32.1	32.5	36.6	44.2	51.5	57.9	62.9	62.4	55.8	48.5	41.7	35.3		
	DBStd	12.28	7.07	6.23	6.36	5.25	5.27	4.42	4.61	4.36	4.50	5.77	5.96	7.33		
	HDD50	2506	556	490	415	186	44	2	0	0	6	98	252	457		
	HDD65	6687	1021	910	880	625	420	217	99	104	277	513	699	922		
	CDD50	1356	0	0	0	12	89	238	399	385	180	50	3	0		
	CDD65	61	0	0	0	0	0	4	32	24	1	0	0	0		
	CDH74	333	0	0	0	0	1	37	210	85	0	0	0	0		
	CDH80	22	0	0	0	0	0	1	15	6	0	0	0	0		
	WSAvg	9.6	11.2	11.3	10.4	9.3	9.3	8.4	7.4	8.3	9.0	9.2	10.4	10.6		
	PrecAvg	26.6	2.1	1.8	1.6	1.4	1.9	2.2	2.4	2.9	2.6	2.6	2.6	2.5		
(15) Precipitation	PrecMax	34.6	4.7	3.8	3.7	2.6	4.2	5.2	7.2	8.5	5.9	5.0	7.6	4.8		
	PrecMin	20.0	0.1	0.4	0.4	0.3	0.4	0.0	0.0	0.5	0.8	0.9	0.4	1.1		
	PrecStd	4.3	1.1	0.7	0.8	0.7	0.9	1.1	1.8	2.1	1.5	1.2	1.4	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.9	47.0	55.5	66.0	71.8	78.2	82.0	80.0	71.7	62.3	55.6	49.5		
		MCWB	46.2	43.8	47.8	53.1	60.2	64.2	68.3	65.3	62.9	57.0	54.1	48.0		
	2%	DB	44.9	44.7	50.9	60.0	68.2	74.1	79.0	76.3	68.5	60.0	53.0	47.8		
		MCWB	43.3	42.4	44.8	50.1	58.7	62.6	66.5	63.9	60.3	56.4	51.2	46.4		
	5%	DB	43.2	42.8	47.4	56.9	64.9	70.4	76.3	73.9	65.9	58.0	51.0	46.3		
		MCWB	41.6	40.9	42.9	48.2	56.0	60.5	64.7	63.1	59.3	54.9	49.4	44.9		
	10%	DB	41.5	40.8	44.7	53.6	61.7	67.8	73.6	71.2	63.5	56.2	49.5	44.4		
		MCWB	40.0	39.0	41.5	46.6	53.9	58.8	63.5	62.7	57.8	53.4	48.0	42.9		
	0.4%	WB	46.5	44.9	49.0	53.9	63.3	67.6	70.8	69.2	64.9	59.3	54.7	48.8		
		MCDB	47.6	45.7	53.7	63.7	69.0	74.8	79.3	76.6	69.9	60.8	55.5	49.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.7	42.8	45.4	51.3	60.1	63.7	68.2	66.8	62.1	57.2	52.0	46.9		
		MCDB	44.6	44.2	48.8	57.9	65.3	70.9	75.7	71.9	66.2	59.1	52.8	47.7		
	5%	WB	41.8	40.9	43.5	49.3	57.7	61.6	66.3	65.3	60.6	55.4	49.8	44.9		
		MCDB	43.0	42.5	46.4	55.2	62.7	68.3	73.4	70.1	63.9	57.3	50.6	46.1		
	10%	WB	40.1	39.1	41.8	47.3	54.9	60.0	64.8	63.9	59.0	53.9	48.2	43.0		
		MCDB	41.4	40.8	44.3	52.6	60.4	66.2	71.4	68.7	62.4	56.0	49.4	44.1		
	5% DB	MDBR	6.7	6.7	10.5	14.2	15.0	15.8	15.3	14.3	12.7	9.6	6.8	6.5		
		MCDBR	6.4	7.2	14.8	19.0	19.5	20.8	20.5	17.8	15.2	10.5	6.8	6.6		
		MCWBR	6.3	5.9	9.6	11.3	11.7	11.1	10.0	8.1	8.6	7.9	6.2	6.6		
		MCWBR	6.5	7.1	13.2	16.7	17.2	19.0	17.5	15.1	12.8	10.4	6.4	6.1		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.4	6.0	9.3	10.8	11.6	11.3	9.7	8.6	8.4	8.4	6.1	6.3		
	taub		0.315	0.338	0.360	0.394	0.391	0.385	0.406	0.407	0.375	0.359	0.339	0.307		
		taud	2.292	2.313	2.304	2.278	2.344	2.388	2.347	2.351	2.420	2.398	2.334	2.269		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		190	231	255	262	270	273	264	255	248	221	179	162		
		Edh at Noon	17	24	31	37	36	35	36	34	28	23	17	14		
	RadAvg		141	335	775	1342	1731	1848	1710	1376	974	476	183	96		
		RadStd	27	55	84	156	172	135	204	100	94	71	32	10		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (59 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-290	N/A	N/A	

Nomenclature: See separate page